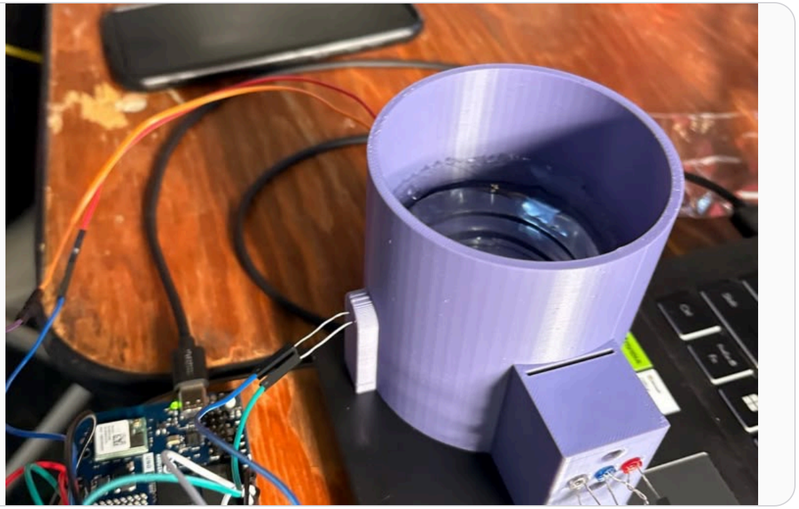
Pill

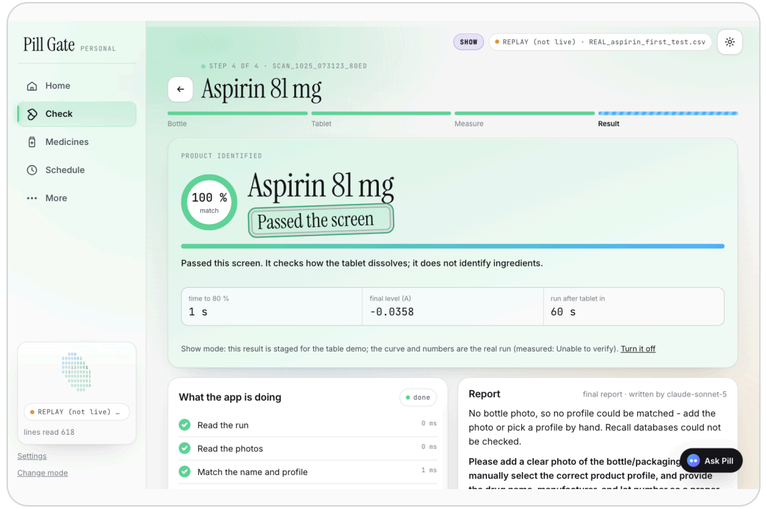
One tablet, one curve, one answer: before the clinic pays the supplier.

A box of hobby parts around an Arduino UNO Q drops one tablet from a medicine delivery into a cup of water, records how it dissolves with two light sensors, and compares the curve with known-good tablets. The station says one of three things, and that answer decides whether the supplier gets paid.

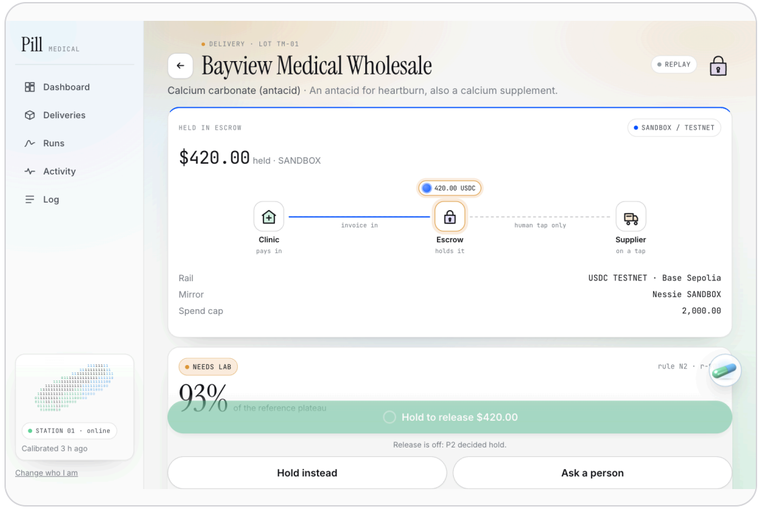
Passed the screen **Needs lab testing** **Unable to verify**



The station. UNO Q, three LEDs, two photoresistors, a printed chamber with a lid.



Station app. A finished run: the curve, the band, the one-line answer.



Clinic app. The verdict holds the payment until a pharmacist taps.

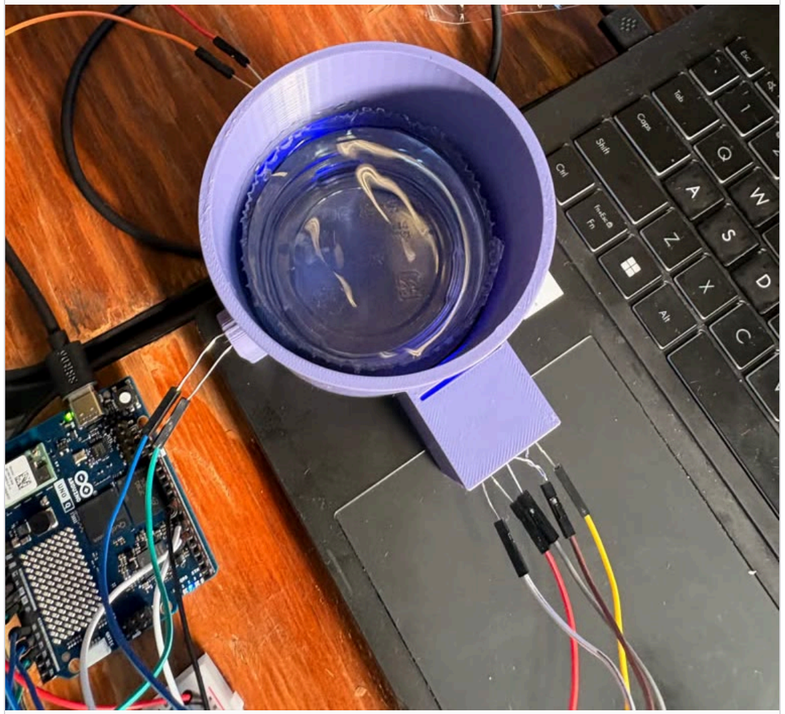
1 in 10
medical products in low- and middle-income countries are substandard or falsified (WHO, 2017)

60 s
from Start to one of three answers; a lab dissolution tester costs $20–100k and takes 7–10 days

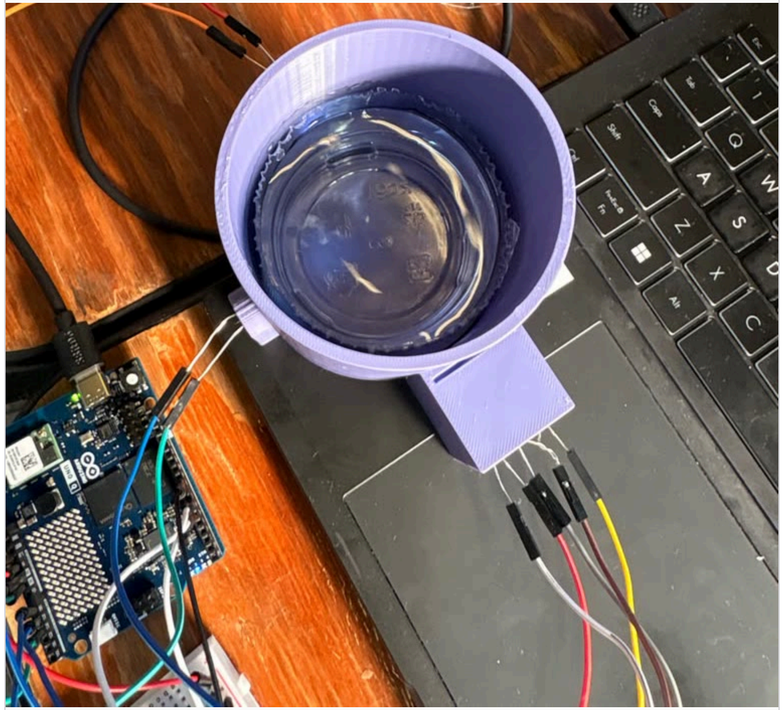
3 answers
never a fourth; no model sets the verdict, code forces the headline to match the measurement

0.05 USDC
per verdict over Coinbase x402 on Base Sepolia (TESTNET); release only after the pharmacist's tap

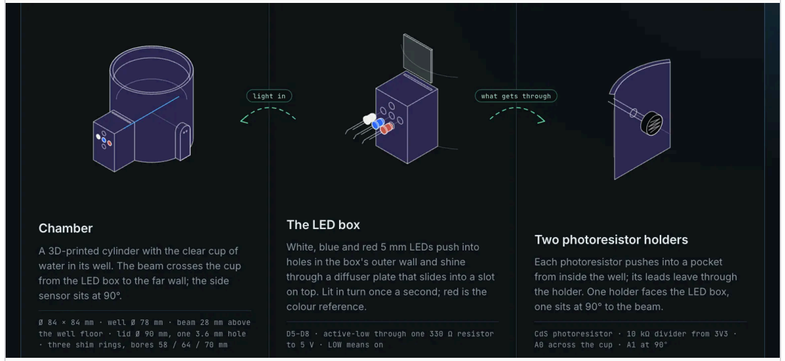
The station: what is in the box



Blue LED on. One photoresistor looks straight through the cup; the other sits at 90° and sees the water go cloudy.



The cup in the chamber. Lid closed; every second the board turns all LEDs off and takes a dark reading, so room light cancels out.

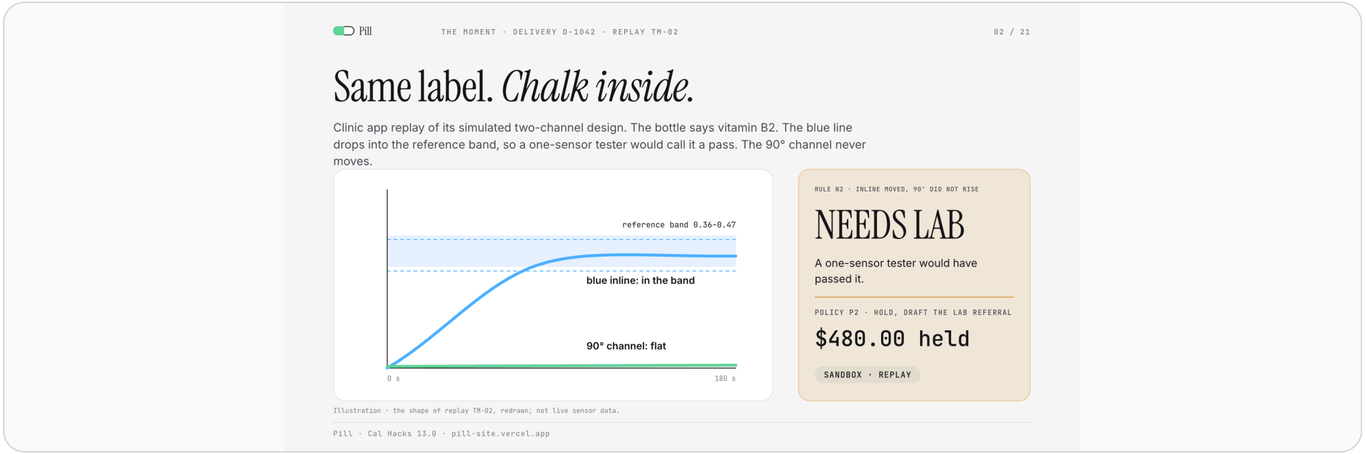


Parts drawing. Arduino UNO Q (3.3 V, 12-bit ADC), two CdS photoresistors, blue / white / red LEDs. The firmware prints one CSV line a second and decides nothing.

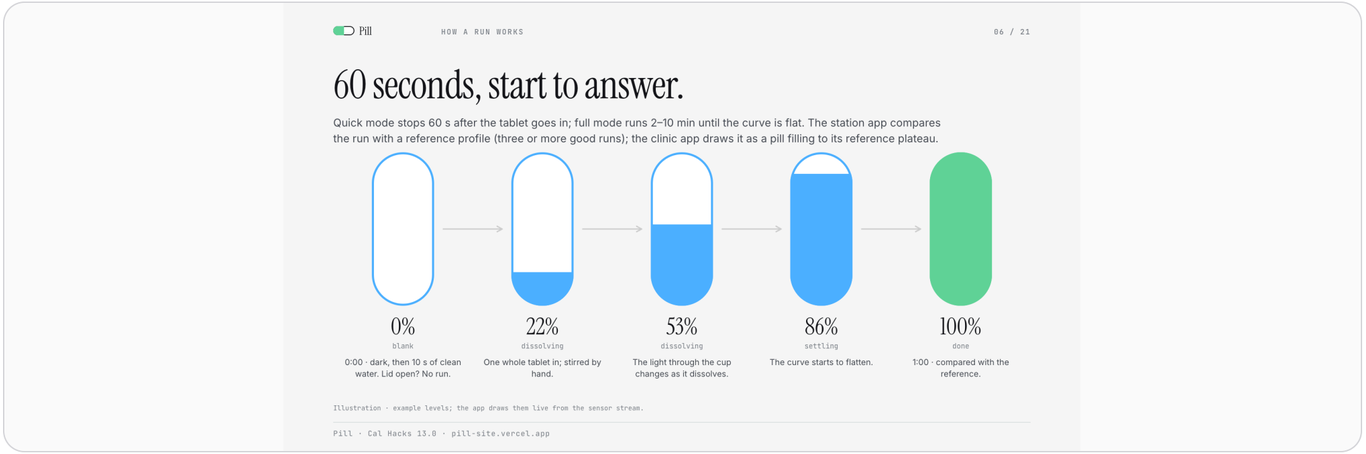
One run: 60 seconds, six phases



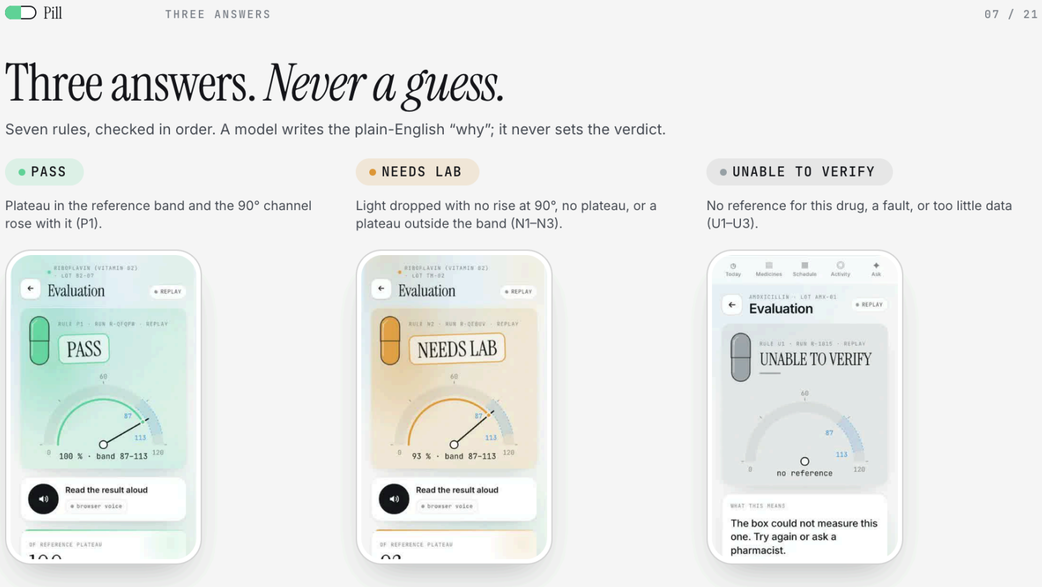
Lights off → clean water (10 s blank) → drop the tablet, 5 stirs, lid → dissolving → settling → verdict. Lid open more than 25 s ends in Unable to verify.



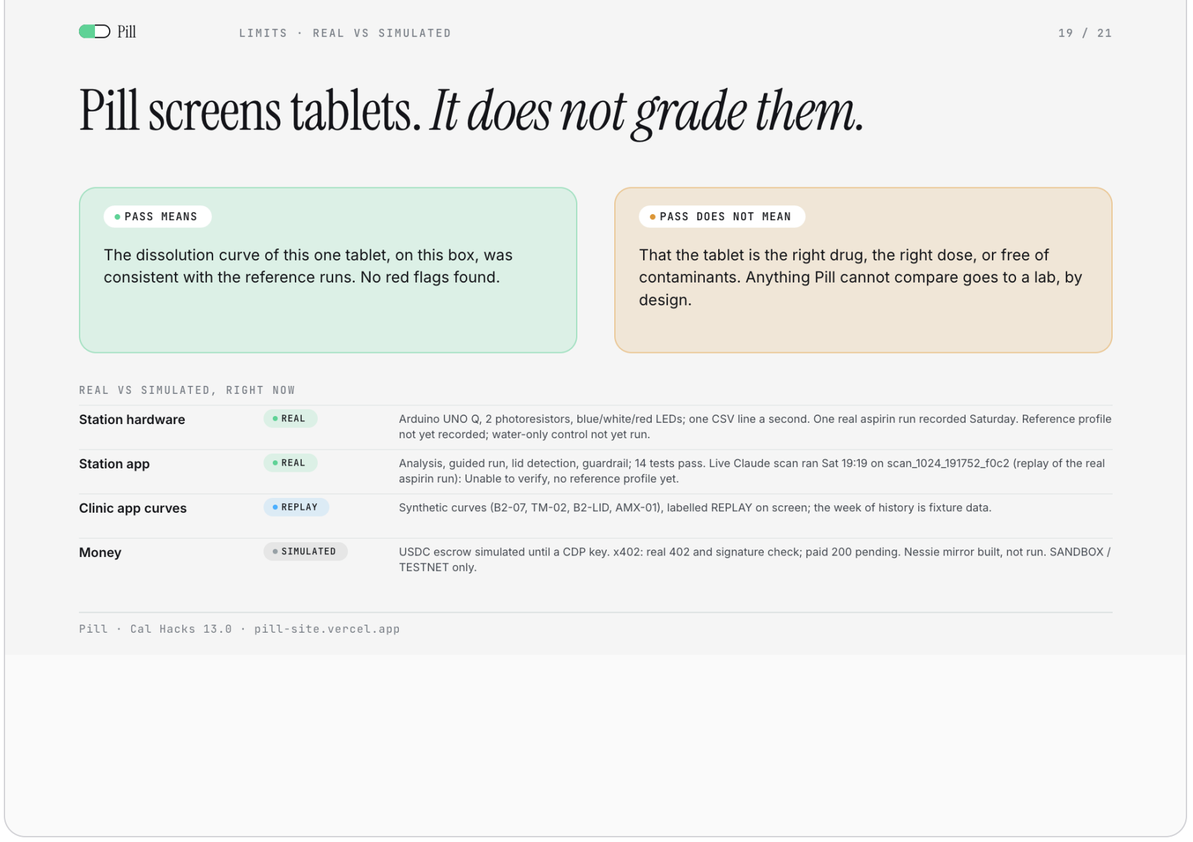
Same label, chalk inside. The 90° channel stays flat where a real tablet clouds the water: Needs lab testing, $480 held (replay of a simulated curve).



Absorbance rises as the tablet dissolves. Aspirin clouds rather than colours the water, so the side sensor carries the answer.

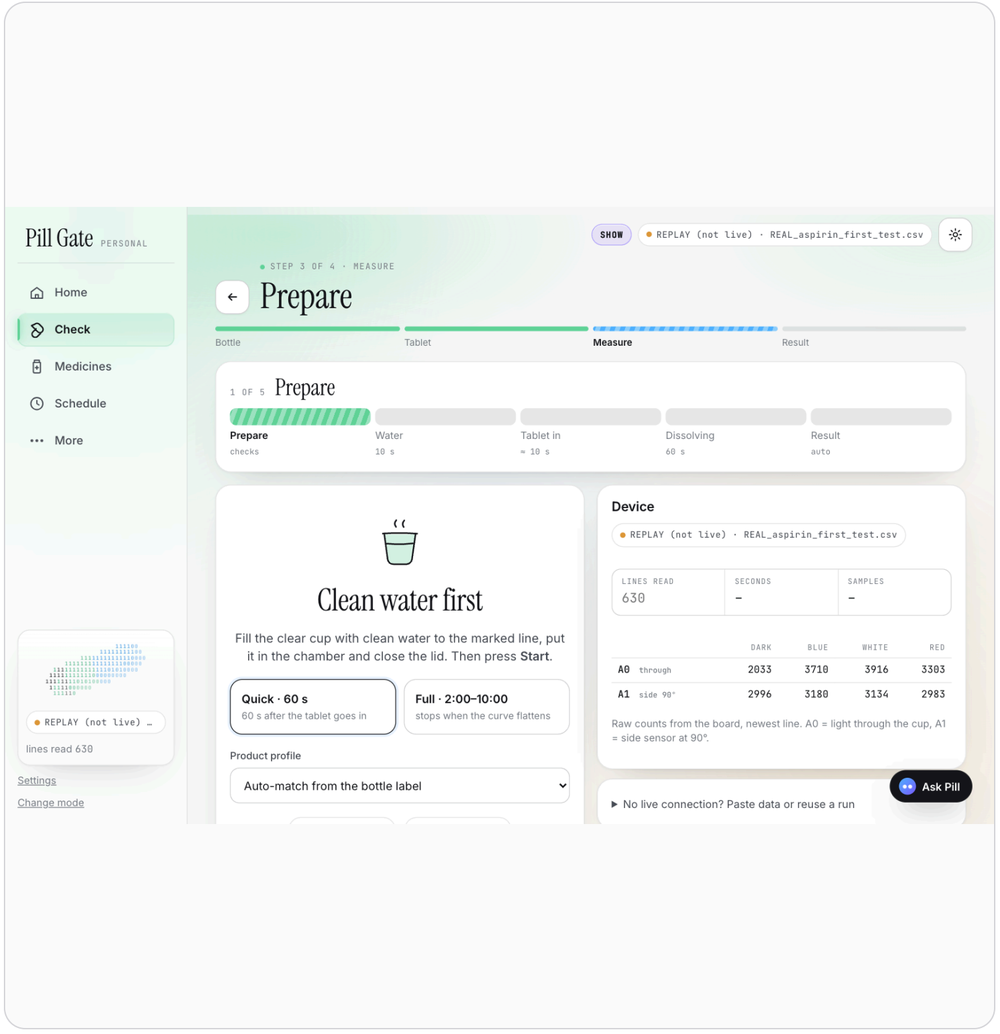


Passed the screen / Needs lab testing / Unable to verify, each with the rule that fired and the plain-English why. A curve cannot prove a medicine is good, so no screen says that.

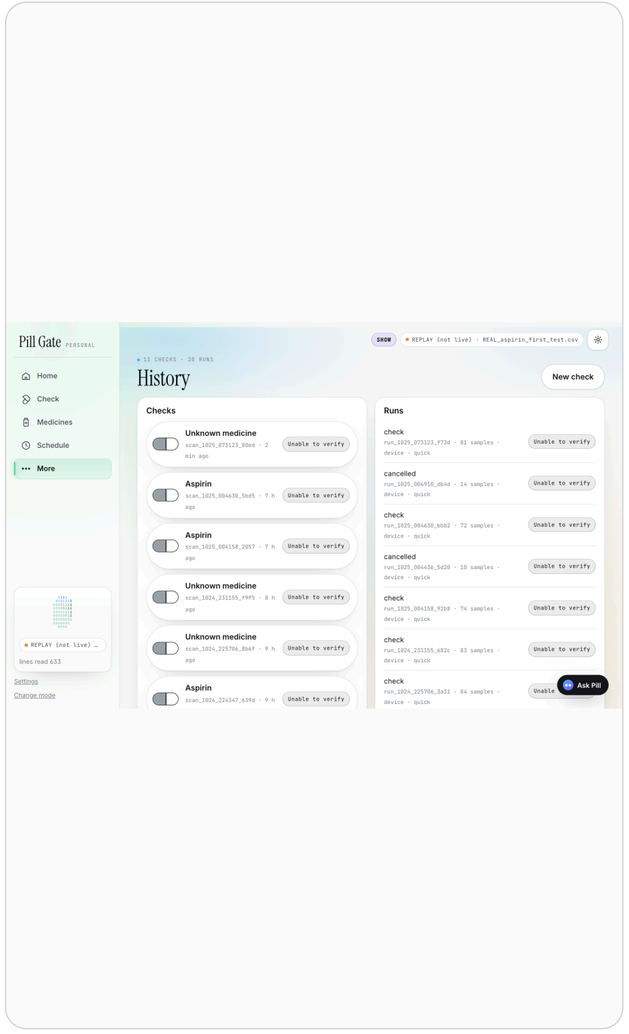


What "Passed the screen" does not mean. It is a screening result, not a lab test. Claude reads the label and the imprint; it never sets the answer.

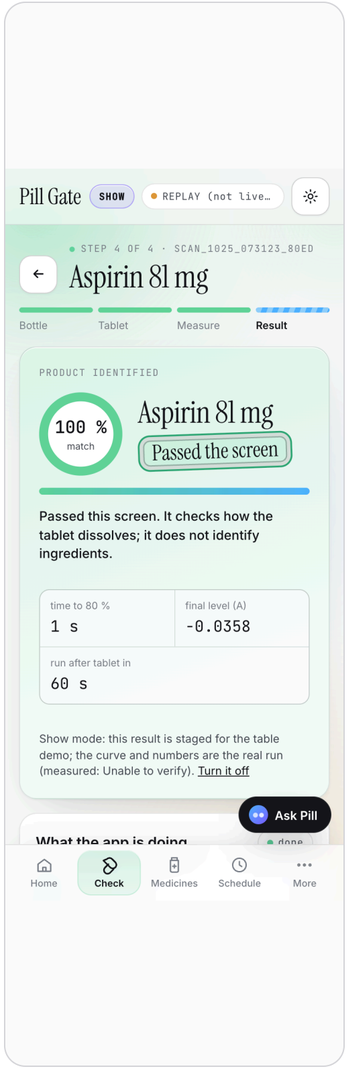
Station app: laptop and phone



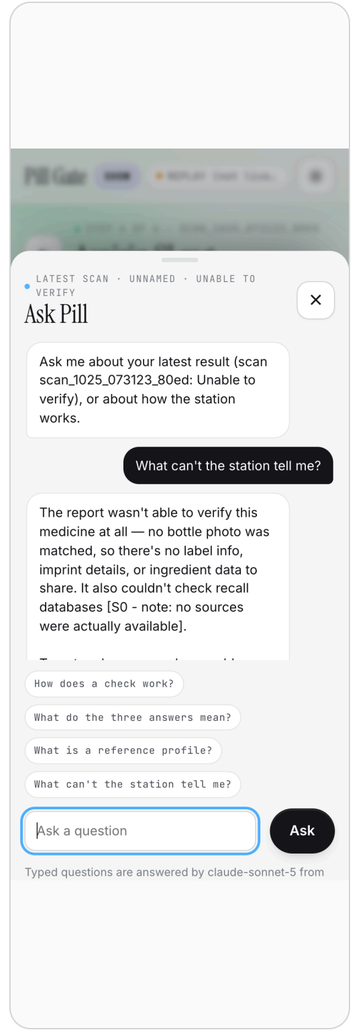
Measure. Prepare → clean water → tablet in → dissolving → result, with the live lines-read counter.



History. Every run with its answer, rule id and curve.

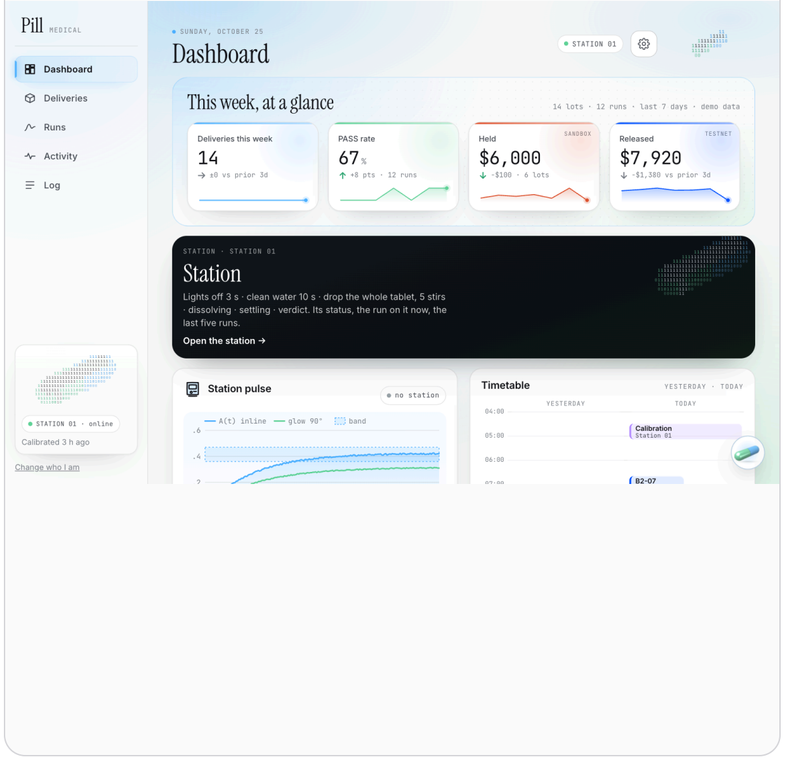


Phone result.

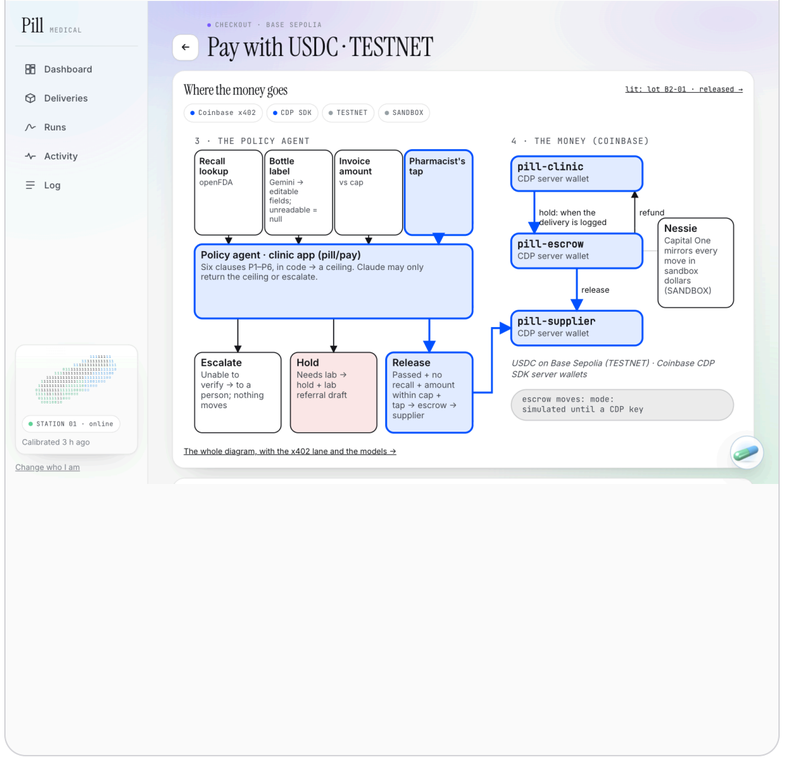


Ask Pill ("Why this answer?").

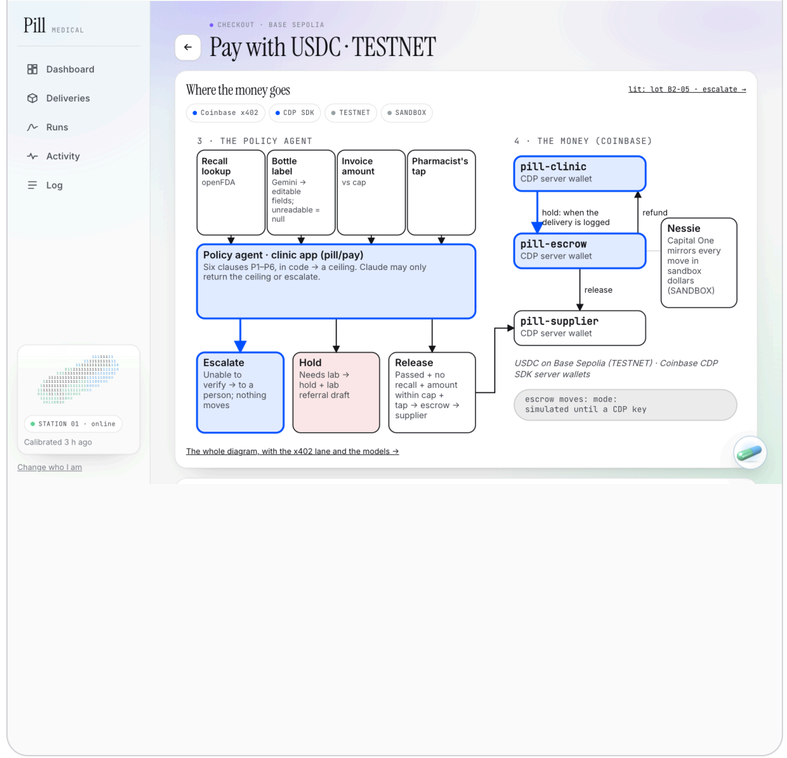
Clinic app: the verdict holds the payment



Dashboard. Deliveries, runs, money held, the day's decisions.

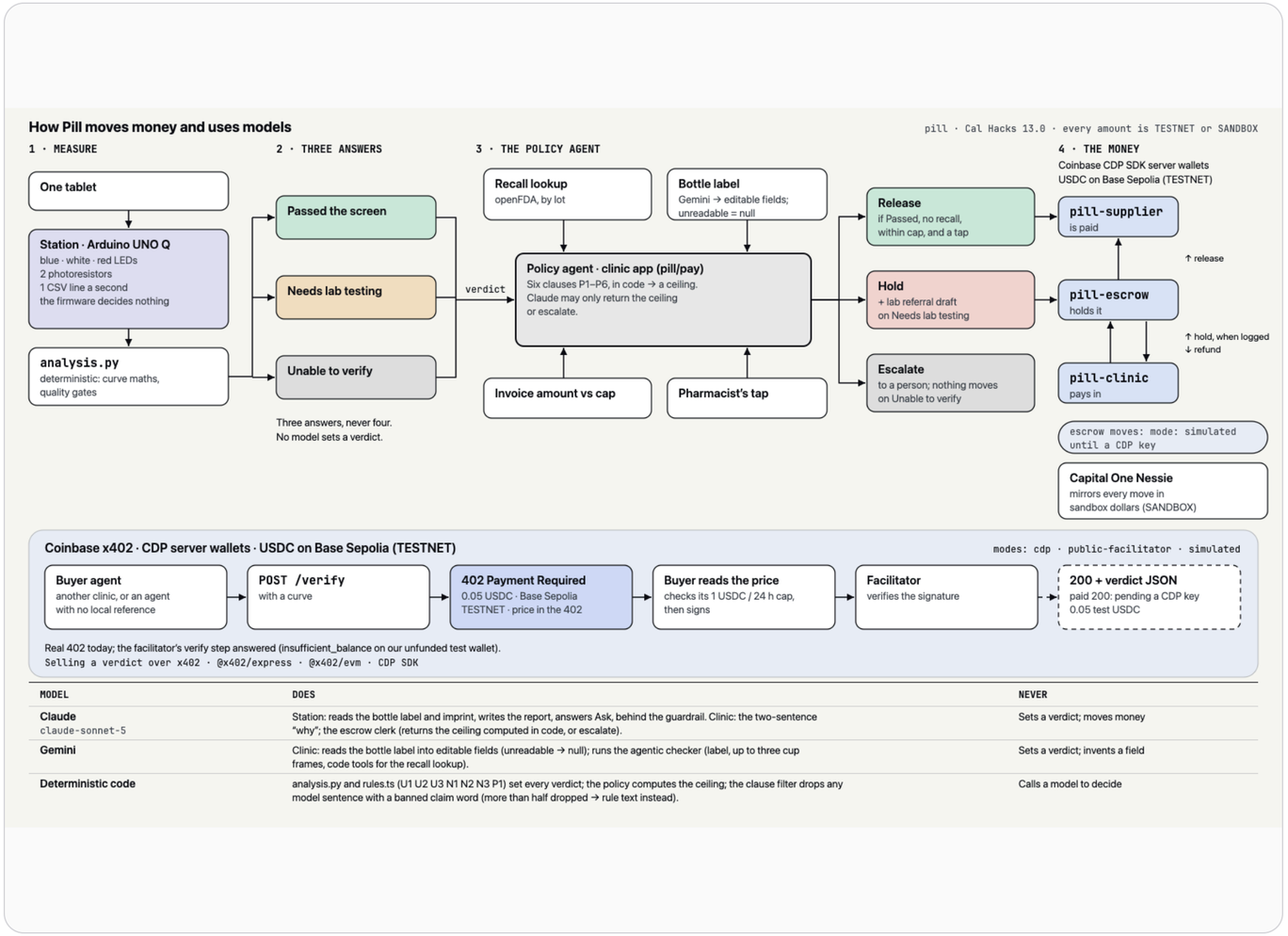


Release. Passed the screen + no recall + within the cap + the pharmacist's tap. Nothing moves without the tap.

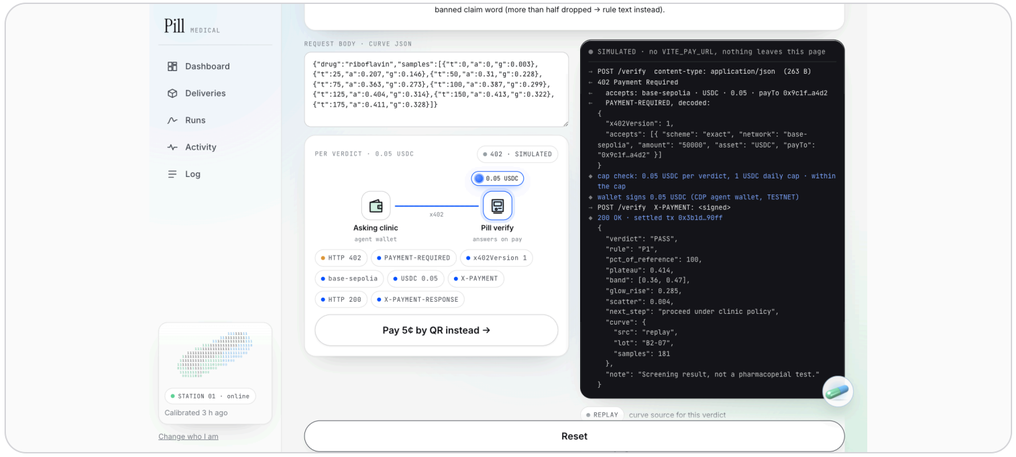


Escalate. Anything but a pass holds the money and quotes the policy clause (P1–P6) that fired.

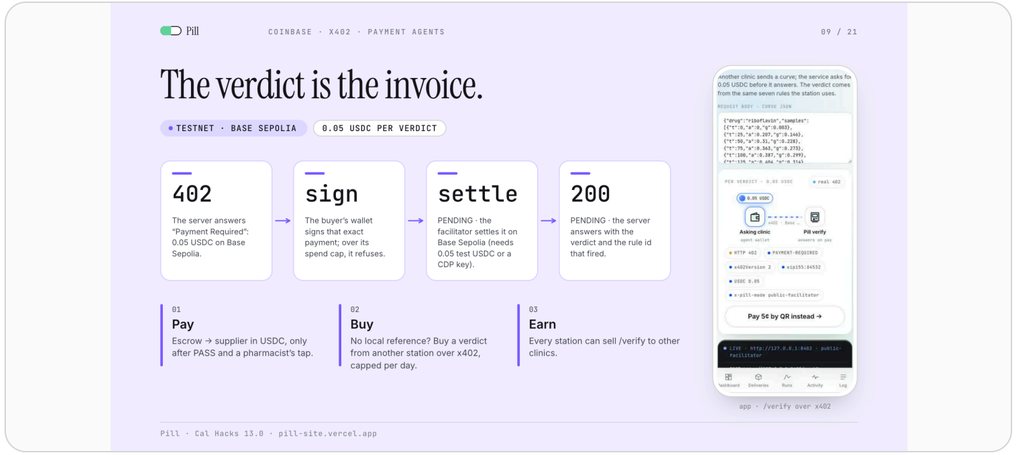
How the money moves: Coinbase x402, CDP wallets, USDC on Base Sepolia (TESTNET)



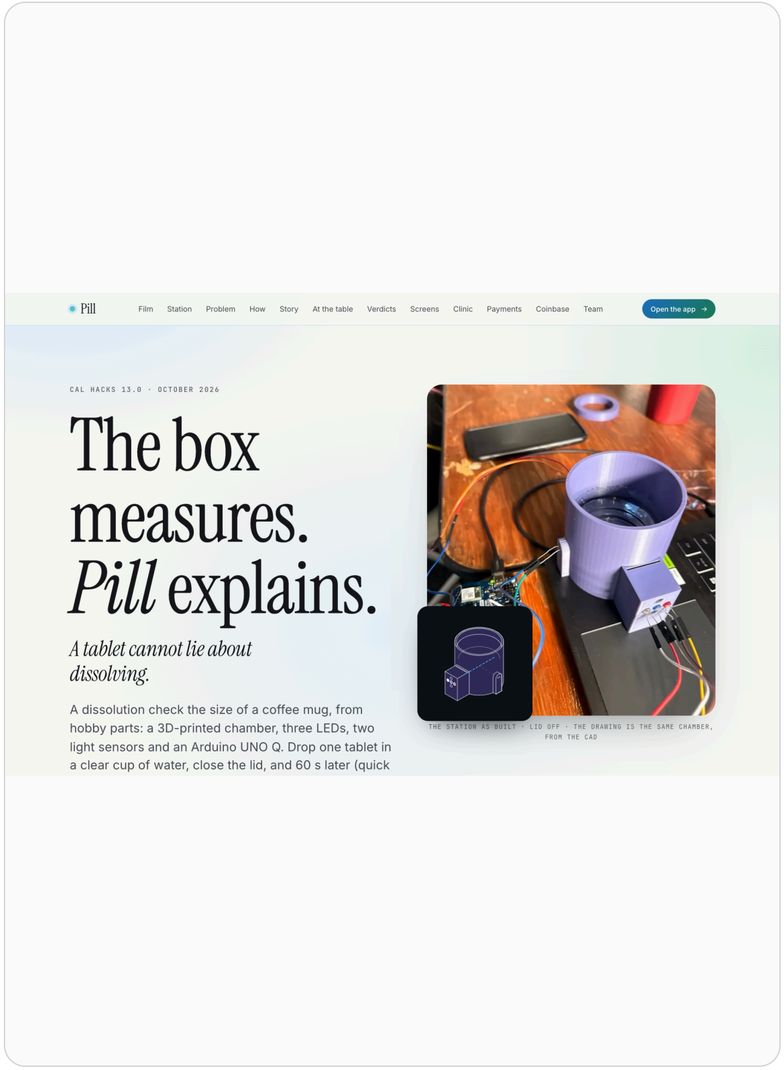
The tree. Measure → three answers → the policy agent → three CDP server wallets (clinic pays in, escrow holds, supplier is paid). Under it, the x402 lane: POST /verify → 402 with the price → the buyer checks its 1 USDC / day cap and signs → facilitator verifies → 200 + verdict. Dashed = not yet run.



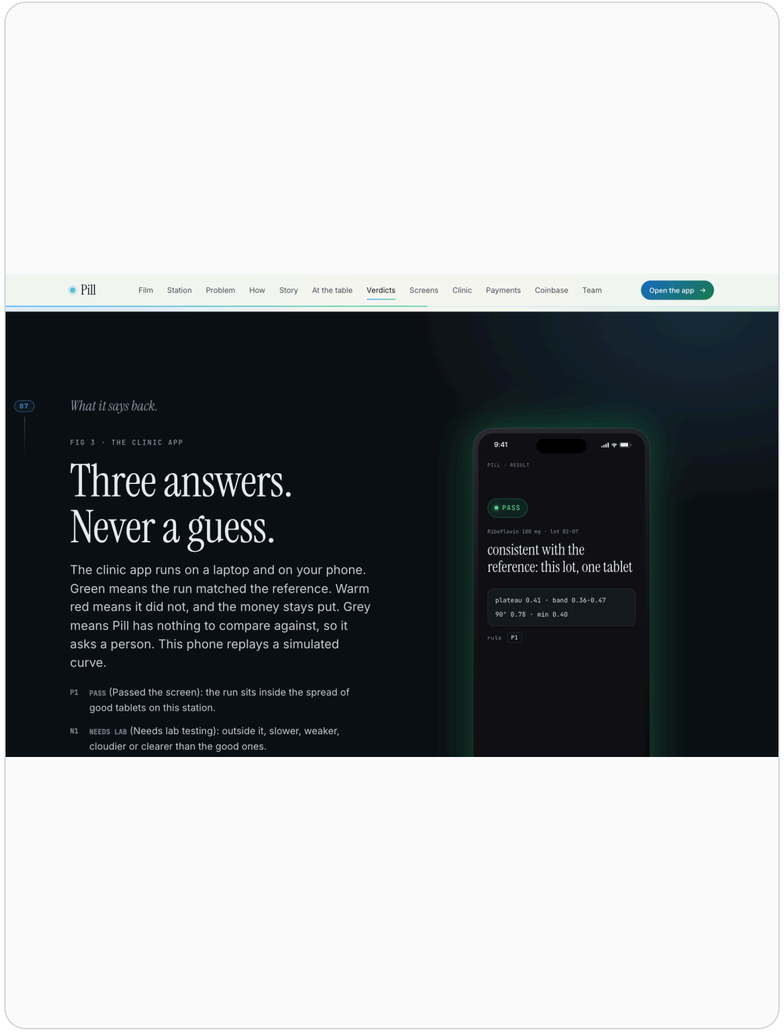
A real 402 today. 0.05 test USDC per verdict; the facilitator's verify step answered for real. The paid 200 is pending a CDP key.



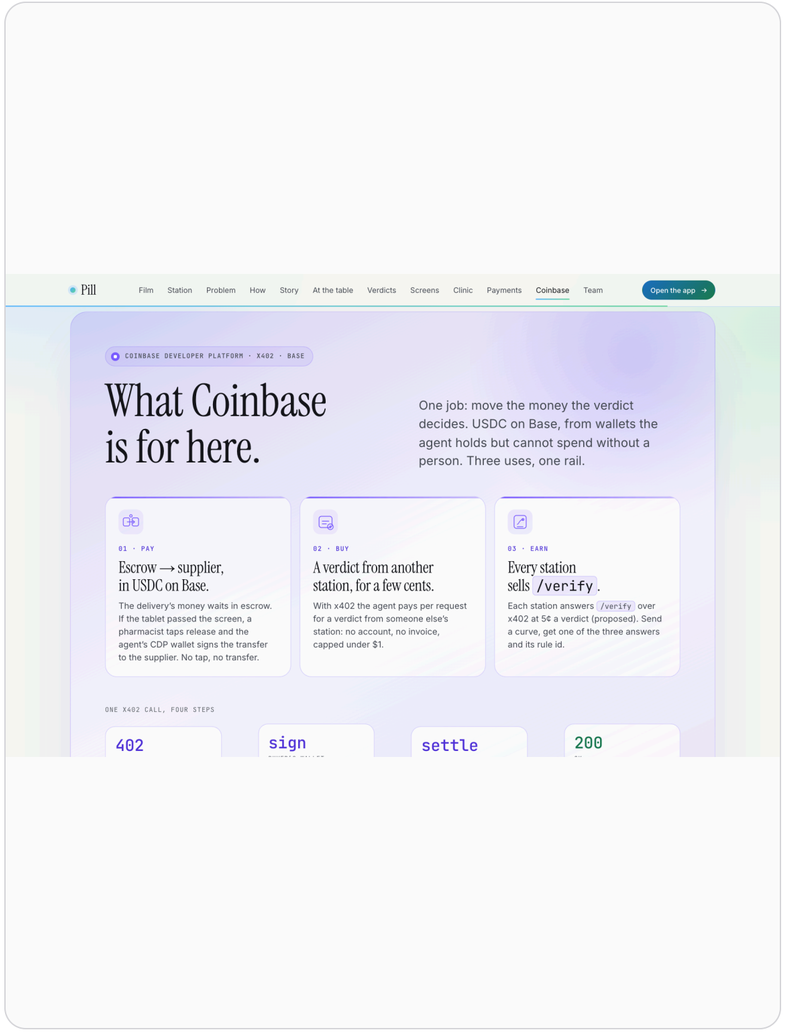
Pay, buy, earn. Escrow → supplier in USDC; buy a verdict from another station for cents; every station sells /verify.



Hero. "The box measures. Pill explains."

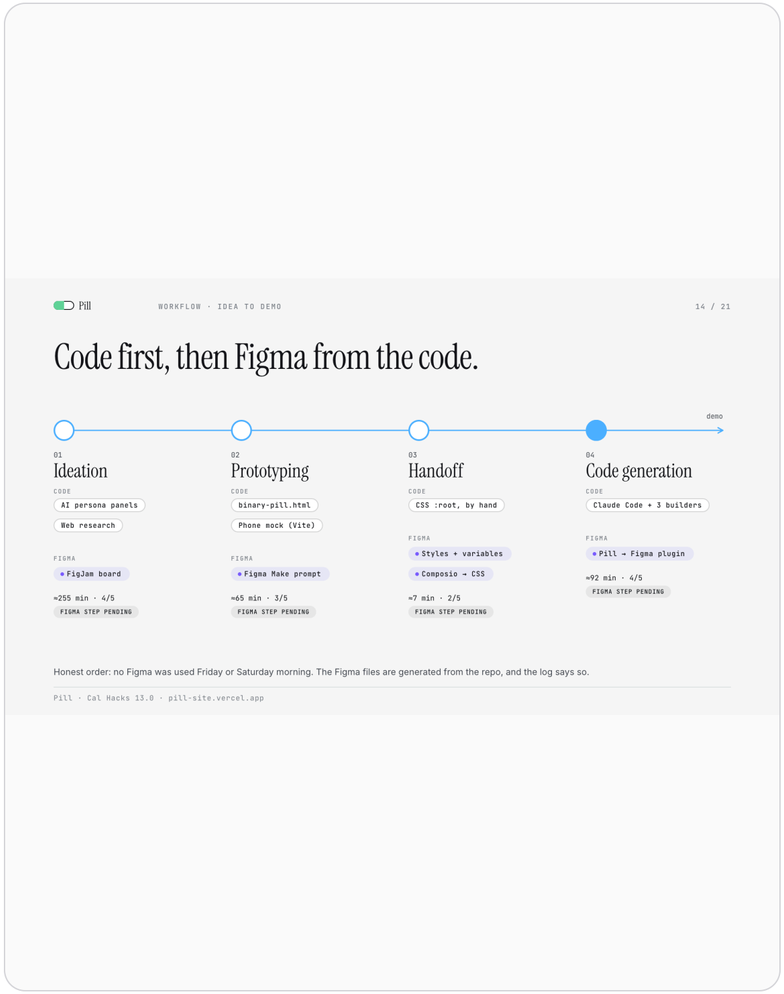


Three answers and what is real now (first aspirin run, no reference yet).

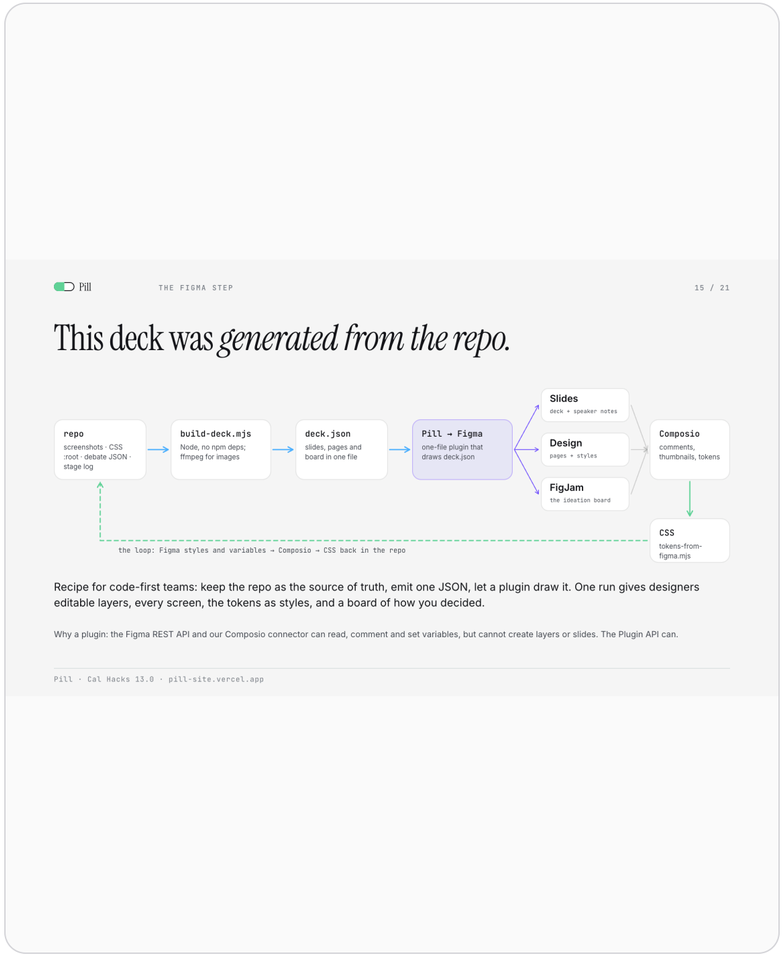


What Coinbase is for here. One x402 call in four steps; press Call /verify to replay 402 → sign → 200.

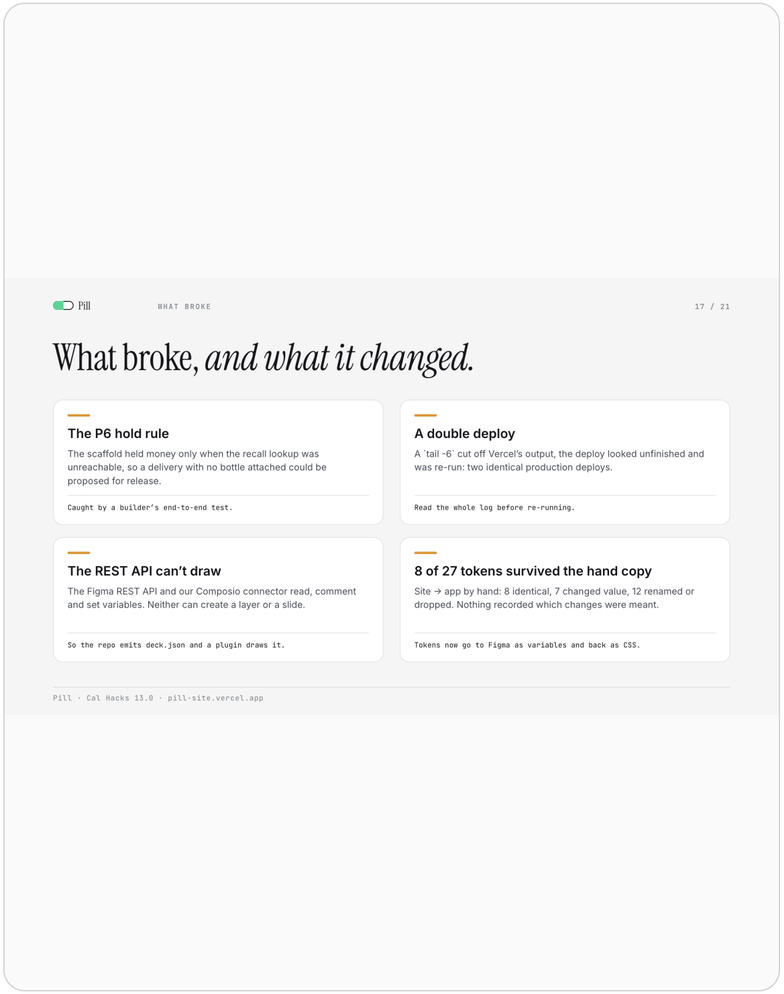
Code first, then Figma from the code



Four stages, timed from git. Ideation with AI judge panels (≈255 min) → prototyping in code + Figma Make (≈65 min) → handoff → code generation.



This deck was generated from the repo. One JSON file → the HTML deck, the PDF, the PPTX, and a Figma plugin that draws the same JSON (tested against a stub of the Figma API).



What broke, and what it changed. The P6 hold rule, a double deploy, the REST API that can't draw, 8 of 27 tokens surviving the hand copy.